

LLM & GENAI

- vLLM
- Unsloth
- LoRA / QLoRA
- PEFT
- Transformers
- LLMOps / Self-hosted Serving
- LLM Evaluation

MACHINE LEARNING

- XGBoost
- Autoencoders
- PyTorch
- TensorFlow
- Keras
- Scikit-learn
- DenseNet
- Logistic Regression
- Classification
- Predictive Modelling
- Unsupervised Learning
- Feature Engineering
- Transfer Learning
- Anomaly Detection
- NLP
- Computer Vision

GPU STACK

- CUDA GPU Libraries
- RAPIDS cuDF / CuPy
- Multi-GPU Training

OPTIMISATION & FORECASTING

- Google OR-Tools
- Vehicle Routing
- Mixed-Integer Programming
- Operations Research
- Constraint Modelling
- Assignment Models
- Supply-Chain Optimization
- Scenario Analysis
- Time-Series Forecasting

PRODUCTION ML PLATFORM

Cloud & Managed ML

- GCP
- GKE
- Vertex AI
- Vertex AI Feature Store
- GCS
- Artifact Registry
- Azure ML
- Function Apps
- Logic Apps

Serving & Orchestration

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

WORK EXPERIENCE

- Lead Artificial Intelligence Engineer

Best Buy · Bengaluru, India

Jul 2025 – Present

 - Built a config-driven **LoRA fine-tuning framework** with Unsloth, PEFT, and Transformers; enabled YAML-driven task addition and single- or multi-GPU training.
 - Design, implement, and maintain end-to-end ML systems spanning data pipelines, training, evaluation, registry, serving, monitoring, controlled releases, and rollback.
 - Deployed **vLLM** serving on GKE with HPA autoscaling, Istio routing, GCS model storage, Artifact Registry, Cloud Build, and Prometheus/Grafana observability.
 - Built fraud systems across batch, streaming, and real-time models. Beam/Dataflow feature pipelines handled **40,000 events/sec** at peak; supported a **200 ms scoring SLA** with secure, monitored GKE/Seldon Core inference and load testing.
- Lead ML Engineer — ML Engineering & Operations

Wipro · Client: Major US Retail Corporation

Aug 2023 – Jul 2025

 - Built behavioural anomaly detection for AWS account security using per-entity autoencoders on **NVIDIA DGX (8×A100)** with NVIDIA Morpheus and billion-row CloudTrail datasets.
 - Developed phishing detection processing **2,400 emails/sec** and RAPIDS data pipelines for GPU-accelerated training and inference; monitored model and system health through MLflow and Grafana.
 - Led ML/MLOps delivery and mentoring at Wipro; completed **300+ résumé screens** and **50+ technical interviews**.
- Senior Data Scientist — Data Science / Azure ML

Accenture

May 2021 – Aug 2023

 - Developed an end-to-end Azure ML product-quality pipeline for BamaGruppen, Norway—**\$2.5M annual savings**, inspections increased from **130 to 160 trucks/day**, and defect detection improved **17%**.
 - Designed Google OR-Tools vehicle-routing models for daily pallets-to-stores delivery under vehicle-count, capacity, time-window, waiting-time, and store-demand constraints; integrated Azure Functions, Logic Apps, and RPA for automated dispatch.
 - Built a Google OR-Tools MIP system for daily transport departures and supply-chain simulations comparing operating cost, product freshness, carbon impact, and regional hub-location scenarios; owned deployment, monitoring, and CI/CD.
- Data Scientist

GD Research Centre Pvt Ltd

May 2020 – May 2021

 - Designed and deployed domain-aware time-series forecasting across **15+ industry verticals**, with feature-engineering pipelines for heterogeneous data and long-horizon business forecasts.

Kubernetes

Seldon Core

Docker

Helm

Istio

Kubernetes HPA

DATA & ML LIFECYCLE

Apache Beam

Dataflow

Pub/Sub

MLflow

Streaming Pipelines

Training Pipelines

Automated Evaluation

Experiment Tracking

Model Registry

Model Versioning

Continuous Training

RELIABILITY, DELIVERY & LEADERSHIP

CI/CD

GitOps

Cloud Build

ArgoCD

GitHub Actions

Prometheus

Grafana

Blue-Green Delivery

Model Monitoring

Rollback

Load Testing

Alerting

Architecture Reviews

Mentoring

Stakeholder Alignment

LANGUAGES, DATA & DATABASES

Python

SQL

Pandas

NumPy

MS SQL Server

IBM DB2

KQL

CERTIFICATIONS +

Oracle Cloud Infrastructure 2024

Generative AI Certified Professional

Microsoft Certified

Azure AI Engineer Associate

Microsoft Certified

Azure Data Scientist Associate

Microsoft Certified

Power BI Data Analyst Associate

EDUCATION

B.E. Electrical & Electronics

Andhra University, Visakhapatnam

2012 – 2016 · 80%

ACHIEVEMENTS

Best Buy Talent Quotient (2026): cybersecurity and fraud delivery.

Deep Learning Engineer

Tata Consultancy Services

Jul 2019 – Mar 2020

► Built and deployed an end-to-end semiconductor ML system: distributed training on ~1 million images in ~3 hours, with 90% classification accuracy. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer — Machine Learning

Infosys

May 2016 – Jul 2019

► Built an ML-based lead-conversion prediction system and business-intelligence applications, following earlier production software work on mainframe systems.

PROJECTS

LLM Fine-tuning & Serving Framework

Best Buy · Jul 2025 – Present

Config-driven LoRA adapter training for open-source causal LLMs with Unsloth and PEFT/Transformers, automated evaluation, GCS artifact lifecycle, and vLLM serving on GKE with autoscaling, Istio routing, and production monitoring.

~\$300K estimated annual API-cost savings saved

Vehicle Routing & Transport Planning

Accenture · May 2021 – Aug 2023

Designed Google OR-Tools VRP and MIP models for daily delivery and terminal-to-distribution-centre planning. Modelled vehicle, capacity, time-window, waiting-time, demand, and departure constraints, then automated operational use through Azure Functions, Logic Apps, RPA, and Slack notifications.

VRP · MIP · Azure Functions

Automated daily transport plans · Replaced manual spreadsheet planning

Supply-Chain Optimization

Accenture · May 2021 – Aug 2023

Built a simulation and optimisation framework covering vendors, warehouses, and customers. Compared operating cost, product freshness, carbon footprint, and network scenarios to identify the appropriate number and locations of regional distribution hubs.

OR-Tools · Simulation · Power BI

Informed regional hub-location and distribution decisions

Industry Forecasting Platform

GD Research Centre · May 2020 – May 2021

Designed and deployed domain-aware time-series forecasting for GlobalData UK across energy, manufacturing, FMCG, healthcare, finance, oil and gas, retail, technology, pharmaceuticals, mining, aerospace, telecommunications, automotive, travel, and other sectors.

Time Series

15+ industry verticals · Enterprise market-intelligence forecasts

Product Quality Prediction

Accenture · BamaGruppen, Norway

Led an end-to-end Azure ML system that predicted product quality at warehouse arrival, automated inspection prioritisation, and integrated model outputs with operational workflows and monitoring.

Azure Pipelines

\$2.5M/year saved · 130 to 160 inspections/day · 17% higher detection

Real-Time Fraud Detection Platform

Best Buy · Jul 2025 – Present

Designed production pipelines from streaming feature generation through model serving, observability, controlled deployment, and rollback for retail fraud decisions across payments, refunds, reshipping, and chargebacks.

40k feature events/sec · 200 ms scoring SLA

ADDITIONAL PROJECTS

- AWS anomaly detection:

per-entity autoencoders on billion-row CloudTrail data with Morpheus, RAPIDS, MLflow, and an 8xA100 DGX.
- Semiconductor defect classification:

DenseNet on about 1 million wafer images, three-hour training cycles, and South Korea deployment.